

KrishokTech: Deterministic-First, Evidence-Bounded Bengali Agricultural Advisory

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Abstract

Bengali agricultural advisory systems face a particular difficulty: treatment requests arrive without a crop name, photographs contradict the accompanying text, and generated dosage claims can exceed what the retrieved evidence supports. KRISHOKTECH treats these uncertainties as explicit control states rather than leaving them to the language model. A deterministic gate halts crop-less treatment queries before retrieval runs. Crop identification fences the evidence set, making off-target passages unreachable, and a confirmation state surfaces when text and image disagree. Before rendering, every dosage-bearing claim in the generated draft is checked against retrieved passages and permitted dose ranges; unsupported claims are dropped. The demonstration follows one farmer interaction through clarification (**ASK**), multimodal conflict (**CONFIRM**), safety referral (**REFER**), and claim filtering (**DROP**). On 200 authentic farmer queries, the gate halted all 30 crop-less requests before retrieval. Crop fencing with contradiction gating reduced off-target retrieval from 36.25% to 28.75% on a 400-query stress set. On the deployed local model, guarded attack success fell from 52.5% to 10.71%; on the cloud benchmark, from 36.19% to 0.95%. The dosage verifier caught 114 of 118 inserted errors in live answers without rejecting any unmodified response.

1 Introduction

When yellow spots appear on potato leaves, a farmer may need advice before an extension visit is available. Phone-based and digital advisory services can extend agricultural information beyond face-to-face extension channels (Department of Agricultural Extension, 2022; Developing Local Extension Capacity (DLEC) Project, 2021; Aker, 2011; Fabregas et al., 2019; Van Campenhout et al., 2021). But in a treatment setting, fluent conversation alone is not enough: an answer for the wrong

crop or an unsupported dosage can turn an incomplete question into a harmful recommendation.

Several small uncertainties can cause this failure. A farmer may ask “which pesticide should I spray?” without naming the crop, leaving retrieval to search across unrelated agricultural guidance. Farmers may also write in colloquial, regional, or Romanized Bangla, while technical guidance uses more formal terminology (Chowdhury et al., 2025; Hossain et al., 2026; Reza et al., 2026a). The photograph and text can disagree as well: the image may indicate one crop while the message names another, and a system that silently chooses one modality leaves the conflict unresolved. Studies of vision-language models show that such conflicts can cause models to over-rely on textual cues or otherwise resolve modalities inconsistently (Deng et al., 2025; Zhu et al., 2024), while Bangla multimodal datasets and benchmarks highlight the additional challenges of low-resource visual-language alignment (Barua et al., 2024; Reza et al., 2026b). Retrieved evidence also does not guarantee faithful generation: language models can produce claims that are unsupported by the supplied context (Mullakkayala et al., 2026). For agricultural advice, this is particularly important for dosage, where a plausible numerical claim can be directly actionable (Reza et al., 2026c).

KRISHOKTECH addresses these failures by making uncertainty and safety decisions explicit rather than leaving them to the language model. The system asks for missing information before retrieval, surfaces disagreements between text and images instead of silently resolving them, refers blocked safety-critical requests to human assistance, and checks generated safety-critical claims before they reach the farmer. These decisions appear as four farmer-visible states: **ASK**, **CONFIRM**, **REFER**, and **DROP**.

Concretely, KrishokTech uses three control mechanisms. First, (**C1**) a deterministic gate halts

underspecified treatment queries before retrieval and requests the missing crop or routes the case to the helpline. Second, (C2) crop identification constrains retrieval to evidence for the identified crop and raises a confirmation state when the farmer’s text and image disagree. Third, (C3) dosage-bearing claims are treated as unverified drafts: each claim is checked against retrieved passages and reference dose ranges from BARI/BIRRI guidance before rendering, and unsupported claims are dropped. Section 5 evaluates each mechanism against the failure it is designed to prevent.

The demonstration follows one continuous farmer session through the four states, from a colloquial crop-less query to crop-scoped evidence, multimodal conflict, safety referral, and verified answer delivery (Section 4). Interactive demo: <https://krishoktech-one.vercel.app>; screencast (<2.5 min): <https://krishoktech-one.vercel.app/screencast>. Source code and installable package are available under Apache-2.0 at <https://anonymous.4open.science/r/KrishokChat-Agricultural-Advisory-System-FCEE>.

2 Related Work and Positioning

Conversational agricultural advisory systems have shown the value of pairing localized agricultural knowledge with a conversational interface. Farmer.Chat combines curated retrieval with structured crop onboarding across multiple languages (Singh et al., 2024). KrishokBondhu targets Bengali-speaking farmers through a voice-first call-centre pipeline that chains speech recognition, retrieval, and generation (Ameen et al., 2026). Krishi Sathi collects intent and slot information over several turns before retrieving (Vijayvargia et al., 2025), and a Bengali RAG system bridges colloquial and formal register through translation and keyword injection (Hossain et al., 2026). Together, these systems establish conversational access, retrieval grounding, multimodal input, and clarification as now-standard ingredients of agricultural advisory NLP.

A second line of work targets evidence grounding and output safety more directly. My Climate CoPilot exposes its intermediate retrieval and reasoning trace for expert review (Nguyen et al., 2025), and AgroLLM checks generated recommendations against structured agronomic thresholds (Ravin-

dran et al., 2026). The underlying risk is not specific to agriculture: vision-language models have been shown to defer to contradictory text over correct visual evidence (Deng et al., 2025), and safety-critical domains such as clinical dialogue face the same tension between fluent generation and verified, bounded advice (Singhal et al., 2023). Work on abstention argues more generally that a system should recognize the limits of its own evidence (Wen et al., 2025; Schönwälder et al., 2025), and clarification-seeking dialogue work shows that a targeted question before answering improves task success and safety (Aliannejadi et al., 2019; Zamani et al., 2023). KrishokTech treats these as one design problem rather than three: the same architecture that decides not to retrieve is also the one that decides not to render.

Most work on retrieval safety, clarification, and prompt-injection defense is developed and evaluated on high-resource, Latin-script English (Fairoze et al., 2025; Jaiswal et al., 2026; Li et al., 2023), and standard Bengali NLP resources remain comparatively scarce and skewed toward formal-register text (Bhattacharjee et al., 2022; Kakwani et al., 2020; Chowdhury et al., 2025; Kabir et al., 2024; Bhowmik et al., 2025). A safety gate tuned only to formal Bengali or English keywords is not a safety gate for the way farmers actually type: colloquial spelling, regional dialect, and Latin-script Banglish all fall outside that coverage (Section 5.3). To our knowledge, KrishokTech is the first system to combine pre-retrieval halting, crop-conditioned evidence fencing, and pre-render dosage verification in one enforceable pipeline for a colloquial, low-resource language (Table 1).

3 System Design

KrishokTech is organized around the four intervention states introduced in Section 1, implemented as five pipeline stages rather than a single retrieve-and-generate step (Figure 1). Each stage narrows what the next stage is allowed to do: **T0** screens the raw request for content that should never reach retrieval or generation (**REFER**); **T1** decides whether the request carries enough information to search at all (**ASK** if not); **T2** constrains what evidence a search may return once it runs, and flags disagreement between modalities (**CONFIRM**); and **T3–T4** treat the generator’s output as an unverified draft until it is checked (**DROP** on failure).

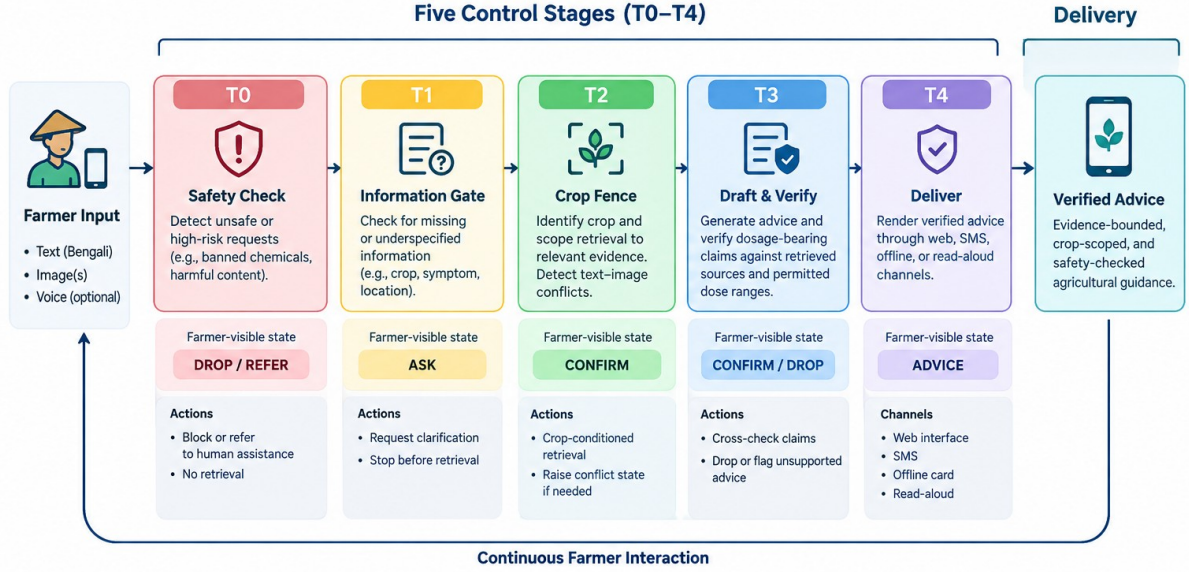


Figure 1: KrishokTech inserts explicit control points before and after generation. Each of the five stages can end a turn in a farmer-visible state: REFER at T0, ASK at T1, CONFIRM at T2, or DROP at T4.

3.1 Halting Before Retrieval

At **T0**, a deterministic precheck matches incoming Bengali queries against banned chemicals (e.g., paraquat), poisoning, and crisis patterns in 0.32 ms and, on a match, **refers** the request to the national 16123 helpline with zero LLM invocation. Requests that clear T0 reach **T1**, where a deterministic extractor pulls crop, symptom, and treatment intent without invoking a language model. A treatment query with no resolvable crop is treated as missing a safety-critical slot: rather than search on an incomplete premise, the pipeline halts before BM25 runs (*sources_retrieved* = 0) and **asks**, offering a non-chemical guidance card, interactive clarification with Bengali quick-reply crop chips, or helpline referral. A *DialectSelector* covers six regional presets to narrow the colloquial-to-formal retrieval gap (Reza et al., 2026a).

3.2 Crop-Fenced Retrieval and the Mismatch Badge

Requests that clear T1 enter **T2**. KrishokTech retrieves via BM25 over 2,135 curated BARI, BRRI, and DAE guideline nodes (Bangladesh Agricultural Research Institute, 2023; Bangladesh Rice Research Institute, 2024; Department of Agricultural Extension, 2022), but only once crop identity is bound; the current deployment covers six staple crops (rice, potato, brassica, chili, corn, wheat), surfaced unobtrusively in the interface. When

a photograph is attached, an on-device ONNX vision model classifies the crop, and a tri-state router sends confident predictions straight to the matching species manual, sends ambiguous pairs to disambiguation chips, and sends out-of-scope images back for re-capture (exact thresholds in Appendix A). The predicted crop then acts as a fence on retrieval: passages for every other crop become unreachable, not merely unranked. When the farmer’s text and the attached image name different crops, KrishokTech does not pick one silently; it raises a **Crop Mismatch** badge and **confirms** with the farmer before releasing any chemical advice. The same discipline applies within a single crop: if the retrieved evidence itself contains conflicting pathogen diagnoses, KrishokTech asks a discriminative question rather than blending the two treatments into one answer.

3.3 Untrusted Generation, Verification, and Delivery

At **T3**, *krishokchat-4b* (a 4-bit quantized Gemma base model fine-tuned with a LoRA adapter, served via *llama-server*) drafts a grounded answer, with *google/gemini-2.5-flash-lite* used as the cloud benchmark throughout evaluation; either way, the draft is treated as unverified text, not a final answer. At **T4**, the dosage verifier checks every numerical claim against a retrieved

Mechanism	Ours	FC	KB	KS	MC
Pre-retrieval halt on missing crop	Yes	Not described	Not described	Partial ^a	Not described
Visual crop-fenced retrieval	Yes	Not described	Not described	Not described	Not described
Pre-render dosage-claim verifier	Yes	Not described	Not described	Not described	Partial ^b
Terminal crisis/poisoning referral	Yes	Not described	Not described	Partial ^c	Not described

Table 1: Control mechanisms as described in the cited papers, not independently tested: Farmer.Chat (FC), KrishokBondhu (KB), Krishi Sathi (KS), and My Climate CoPilot (MC). “Not described” does not mean the system lacks the mechanism. ^aCollects missing slots over multiple turns but does not halt retrieval specifically on an unresolved crop. ^bExposes an evidence trace for expert review rather than an automatic pre-render check. ^cEscalates uncertain calls to a human operator, without a dedicated crisis or poisoning branch.

passage and a permitted BARI/BRRI dose band: a claim that is not bound to evidence, or that falls outside the permitted range, is annotated and **dropped** rather than shown. Surviving responses carry one of five provenance badges (fact lookup, grounded generation, guidance card, clarification, or referral; detailed in Appendix A), so a farmer can see, at a glance, why a given answer is on screen. The verified decision then reaches the farmer through one of four channels: an interactive web advisory, a 160-character SMS, an offline cached card served from local BM25, or sentence-synchronized Bengali neural read-aloud.

4 Demonstration

The demonstration is intended for researchers and practitioners who build multilingual, multimodal, or safety-critical conversational systems. Attendees follow one farmer session on the live system in five steps, S1 to S5, and the screencast follows the same order. The steps cover all four states from Section 1: **ASK** (S1), **CONFIRM** (S3), **REFER** (S4), and **DROP** (S5). S2 shows the crop fence that S3 depends on. Attendees can start from the sample queries on the start screen, which include regional-dialect and safety-test sets, or enter their own queries and photographs.

S1: Missing crop (ASK). The farmer types a colloquial query that names no crop: “*Patay holud dag hoyeche, ki bish dibo?*” (“Yellow spots have appeared on the leaves. Which pesticide should I use?”). A treatment request without a crop lacks a safety-critical detail, so the **T1** gate stops the request before retrieval (*sources_retrieved* = 0) and asks which crop is affected. The reply shows crop chips (Figure 2a). The farmer taps *Alu* (Potato), which binds the crop, and the system answers from potato guidance with numbered citations.

S2: Photograph scopes the evidence. The

farmer then uploads a leaf photograph on the diagnosis page. The on-device classifier identifies the crop as rice and runs the rice disease model. The chat that opens from this page takes rice as its context, and the **T2** fence lets retrieval return only rice guidance. The photograph never leaves the device.

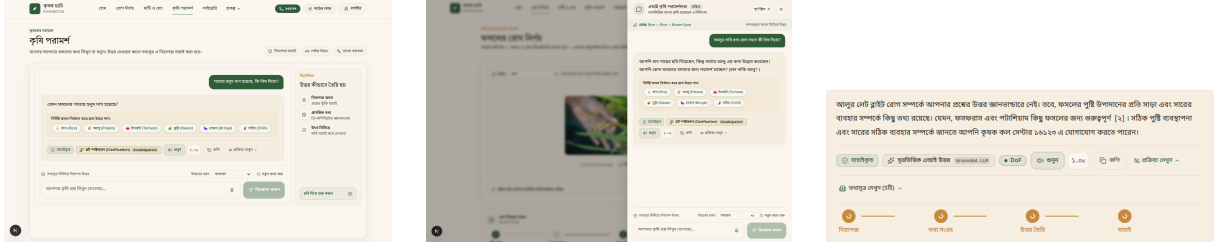
S3: Photograph and text disagree (CONFIRM). With the rice photograph still attached, the farmer asks how to control potato late blight. The system does not pick a crop. It raises a **Crop Mismatch** badge, states that the photograph shows rice while the message names potato, and withholds chemical advice until the farmer confirms the crop (Figure 2b).

S4: Banned pesticide (REFER). The farmer next asks how to spray paraquat, a banned pesticide. The **T0** precheck matches the request before any retrieval or generation, so the system produces no advice. It shows a banned-chemical warning and a button to call the Krishi Call Centre (**16123**).

S5: Verified answer (DROP). The farmer finally asks which fungicide to spray against potato late blight. The system retrieves evidence, drafts a Bengali answer, and checks each dosage sentence against a retrieved passage and a permitted dose range. In the demonstrated case, the **T4** verifier drops one sentence. The farmer sees the remaining answer with its citations and a provenance badge, and the Why panel shows the verification trace behind that decision (Figure 2c). The same verified answer can be read aloud in Bengali, formatted as a 160-character SMS, or opened as an offline card. Appendix E shows the screens for each step.

5 Evaluation

We evaluate each control against the failure it is meant to prevent. Table 2 gives the headline results, and Appendix F lists every measured result with its status.



(a) S1 (ASK): no crop named, so the system asks and offers crop chips.

(b) S3 (CONFIRM): the photo shows rice, the message names potato, and the system asks which is meant.

(c) S5 (DROP): the Why panel shows the verification trace behind the delivered answer.

Figure 2: Live interface at three demonstration steps. Appendix E shows the other steps.

5.1 Pre-Retrieval Gate (C1)

A treatment request that names no crop cannot be answered safely, because open retrieval may return guidance for a different crop. We test the gate on 200 authentic Bengali farmer treatment queries, 30 of which name no crop. Two annotators labeled the crop slot independently and agreed on every crop-presence decision (Cohen’s $\kappa = 1.00$). The gate halted all 30 crop-less queries before retrieval, so these turns retrieved no sources, and it passed the other 170 to fenced search without a false halt. An unconstrained LLM asked to make the same decision halted 10% of the 200 queries, against 15% for the gate, agreed with the gate on 91% of decisions, and took a median of 4.2 s per query. On this set, a deterministic slot check matched the annotators on every query without calling a model. Slot-extraction accuracy, the controlled pilot, and token metering are in Appendix A.

5.2 Crop Fence and Mismatch Badge (C2)

Retrieval that ignores the crop can return guidance for the wrong crop, and a photograph can contradict the farmer’s text. The six on-device disease classifiers reach 0.94 to 0.99 top-1 accuracy on 4,294 held-out test images. To test the fence, we use a 400-query stress set of dialectal, ambiguous, and conflicting queries, chosen because open search drifts on them. A query yields wrong-crop advice when its top five sources include a document for another crop. Open retrieval does so for 36.25% of the queries. Fencing retrieval to the crop predicted by the on-device vision model, at its empirical error rate of about 5%, lowers this to 31.25%. Adding contradiction gating, which halts conflicting queries before retrieval, lowers it further to 28.75%.

The mismatch badge caught 453 of 454 explicit text-image contradictions (53 of 54 farmer

cases and all 400 cases in PRISM, our synthetic multimodal-conflict suite) with no false halts. The one miss used an implicit crop reference. On a separate set of 300 conflict queries, a prompted LLM asked the farmer to clarify in only 20% of cases, and naive multimodal RAG in 13%. The fence and badge therefore reduce wrong-crop advice on a set where open search drifts, but they do not remove it. Per-crop accuracy, quantization, and the full fence comparison are in Appendix B.

5.3 Safety Verification and Constrained Delivery (C3)

We first measure attack success rate (ASR), the share of adversarial prompts that yield an unsafe response, with the deterministic controls on (guarded) and off (unguarded). Across 420 live calls to the cloud benchmark model (gemini-2.5-flash-lite), 210 per arm, the guarded pipeline had an ASR of 0.95% against 36.19% unguarded. On the Bengali-native subset of 100 attacks, the guarded ASR was 10.0% against 83.0%, with the residual failures concentrated in embedded (4 of 10 prompts) and role-play (2 of 10) variants. On the deployed model, krishokchat-4b (280 calls per arm), ASR falls from 52.5% to 10.71% overall and from 77.0% to 19.0% on the Bengali-native subset, so the residual ASR is higher than on the cloud model. Latin-script Banglish and phonetic spellings such as *parakwat* also evade keyword gates written for formal text. A gate limited to formal English and Bengali script intercepted 1 of 85 adversarial queries in this style, whereas our deterministic precheck intercepted all 85 and raised no false alarm on 15 benign farmer questions.

Next we look at the advice itself. In a blinded audit, three evaluators (two agronomy or botany undergraduates and one extension officer) rated 46

Control	Test	Result
Pre-retrieval gate (C1)	200 farmer treatment queries, 30 without a crop	All 30 halted before retrieval; no false halts
Crop fence and mismatch badge (C2)	400 stress queries; 454 text-image contradictions	Wrong-crop advice 36.25% $\rightarrow$ 28.75%; 453 of 454 contradictions caught
Attack guard (C3)	420 cloud-model and 560 deployed-model calls, half guarded	Attack success 36.19% $\rightarrow$ 0.95% (cloud); 52.5% $\rightarrow$ 10.71% (deployed)
Dosage verifier (C3)	118 inserted errors in 38 live answers	114 caught; 0 of 38 unmodified answers rejected

Table 2: Headline results. Arrows show the rate without and with the control. Appendix F gives sample sizes, confidence intervals, and the scope of each result.

of 47 guarded outputs as safe and actionable, one as vague but harmless, and none as dangerous (Fleiss’ $\kappa = 0.82$). All three unguarded control outputs were rated dangerous. For the dosage verifier, we inserted controlled errors into verified answers. On 38 live answers with 118 inserted errors, it caught 114 and rejected none of the unmodified answers. On a scaled benchmark of 528 cases from 88 BARI and BRRI passages, it caught 296 of 350 inserted errors (84.6%): every overdose (76 of 76) and underdose (67 of 67), but only 85% of chemical substitutions, 73% of unit substitutions, and 65% of invented pre-harvest intervals.

Constrained channels must also preserve the dose. The SMS packer kept 399 of 400 messages within 160 characters. On 100 long advisories that carry a dose, naive truncation preserved the dose in 23, an LLM summarizer in 89 (with one invented dosage), and our deterministic compressor in 92, without length violations or invented values. Local BM25 retrieval, which serves the offline card, had a 0.94 hit rate and was unaffected by simulated packet loss, while remote retrieval lost 13 to 30 percentage points. The non-LLM stages add a median of 33.6 ms per request. Appendices C and D give the attack families, audit protocol, mutation table, and delivery measurements.

5.4 What This Evaluation Does Not Establish

The results hold only under the conditions in which they were measured. We ran no field study, so farmer adoption and yield effects are unmeasured. The crop-fence results come from a benchmark of paired queries under empirical vision noise, not from live farmer sessions, and most contradiction cases (400 of 454) are synthetic. The expert audit covers 47 guarded outputs, network resilience was tested only with simulated packet loss, and the dosage verifier checks numerical dosage statements, so other agronomic claims in an answer

are not verified. Appendix F marks each result as measured, simulated, pilot, or modeled.

6 Limitations, Availability, and Conclusion

6.1 Limitations

Four limitations matter most; Appendix F gives the full ledger.

- Coverage is limited to six staple crops and Bengali input. Retrieval is lexical (BM25), and dense retrieval was not evaluated.
- The mismatch badge detects explicit crop conflicts only, and it missed the one implicit reference in our test set.
- The safety controls lower attack success but do not remove it, and the deployed local model is more exposed than the cloud model (10.71% against 0.95% guarded attack success).
- SMS compression is lossy: on 100 long advisories it kept the dose in 92 but the chemical name in only 34. Generation with the local model on CPU takes about 24 s per call.

6.2 Availability and Licensing

KrishokTech is live at <https://krishoktech-one.vercel.app> (screen-cast: <https://krishoktech-one.vercel.app/screencast>).

6.3 Conclusion

KrishokTech makes the boundary between model output and verified evidence visible to the farmer. Instead of leaving uncertainty and safety to the language model, it asks for a missing crop, confirms conflicts between text and photograph, refers unsafe requests to a human helpline, and drops dosage

claims that the evidence does not support. Our evaluation shows that these controls reduce wrong-crop advice and unsafe output without eliminating either, so the system is meant to support extension officers and not replace them. A field study with farmers is the natural next step.

Ethics and Broader Impact Statement

Agricultural advice for smallholder farmers affects food security, livelihoods, and physical safety. Incorrect chemical or dosage guidance is the central risk this paper is designed around.

Farmer Query Data. The 200 farmer queries in the gate evaluation (Section 5.1) come from extension-service interaction logs. Names, phone numbers, and plot locations were removed before analysis. The queries were collected in collaboration with regional extension officers under institutional review at North South University (Institutional Review Board / Research Ethics Committee), and informed consent was obtained for non-commercial research and safety evaluation. Two annotators labeled the crop slot independently, and consensus adjudication set the final labels. Agreement was 100.0% for crop presence (Cohen’s $\kappa = 1.0000$), 99.5% for the normalized 37-crop taxonomy ($\kappa = 0.9945$), and 93.5% for query intent ($\kappa = 0.8958$).

Agricultural Image Data. Farmer leaf photographs are classified on the device (WebAssembly and ONNX) and are never uploaded or stored remotely, which is a deliberate privacy choice. The held-out images used to measure vision accuracy (Section 5.2) are separate from farmer photographs. They come from open-access agricultural datasets: PlantVillage, the Bangladesh Rice Knowledge Bank, and Kaggle agricultural datasets under CC-BY and open research licenses.

Pesticide and Dosage Guidance. The chemical-advice pathway fails closed. Banned active ingredients and acute-poisoning language trigger a deterministic block and a helpline referral before any generation, and dosage claims in generated text are checked against source passages and permitted dose bands before display (Section 3.3). These controls reduce risk but do not remove it. Guarded attack success is 10.71% on the deployed model (19.0% on Bengali-native attacks), and the dosage verifier missed 4 of 118 inserted errors (Section 5.3). We report these residuals and do not claim that the system prevents all unsafe output.

Helpline Referral. Unsafe and crisis-pattern requests are referred to the national Krishi Call Centre (16123), a government agricultural advisory service. The centre is open from 9:00 AM to 5:00 PM, Saturday through Thursday, except on public holi-

days. The referral screen displays these hours and tells off-hours users to wait for the next shift or seek in-person extension help. The public demonstration runs in a sandbox that shows referral and safety blocking for toxic-chemical queries without giving any chemical advice.

Third-Party Model Use. The cloud benchmark uses a third-party API (google/gemini-2.5-flash-lite) under commercial enterprise terms with a zero-data-retention agreement, so prompts and responses are not logged, retained, or used for training. Cloud calls were limited to offline benchmark evaluation and never received farmer images or direct personal identifiers. De-identified farmer text queries, with personal identifiers removed as described above, were also sent to this API for the live-model gate comparison in Section 5.1. Self-hosted deployment runs entirely on the local `krishokchat-4b` model through `llama-server` and makes no external network calls.

Human Oversight. KrishokTech is meant to support extension officers and not to replace them. Unresolved or unsafe cases end in a referral to a human expert instead of a generated answer. All results are scoped to the conditions in which they were measured (Section 5.4), and simulated or bounded components are disclosed as such and not presented as field-validated.

References

- Jenny C. Aker. 2011. [Dial “a” for agriculture: a review of information and communication technologies for agricultural extension in developing countries](#). *Agricultural Economics*, 42(6):631–647.
- Mohammad Aliannejadi, Hamed Zamani, Fabio Crestani, and W. Bruce Croft. 2019. [Asking clarifying questions in open-domain information-seeking conversations](#). In *Proceedings of the 42nd International ACM SIGIR Conference on Research and Development in Information Retrieval*, pages 475–484.
- Mohd Ruhul Ameen, Akif Islam, Farjana Aktar, and M. Saifuzzaman Rafat. 2026. [KrishokBondhu: A retrieval-augmented voice-based agricultural advisory call center for bengali farmers](#). In *2026 IEEE International Conference on Quantum Photonics, AI, and Networking (QPAIN)*.
- Bangladesh Agricultural Research Institute. 2023. [Krishi projukti hatboi \(handbook on agricultural technology\)](#). *BARI Publications, Gazipur, Bangladesh*.

- Bangladesh Rice Research Institute. 2024. Adhunik dhaner chas (modern rice cultivation manual). *BRRRI Publications, Gazipur, Bangladesh*.
- Deeparghya Dutta Barua, Md Sakib Ul Rahman Sourove, Md Fahim, Fabiha Haider, Fariha Tanjim Shifat, Md Tasmim Rahman Adib, Anam Borhan Uddin, Md Farhan Ishmam, and Md Farhad Alam. 2024. [Chitrojera: A regionally relevant visual question answering dataset for bangla](#). *arXiv preprint arXiv:2410.14991*.
- Abhik Bhattacharjee, Tahmid Hasan, Wasi Ahmad, Kazi Samin Mubasshir, Md Saiful Islam, Anindya Iqbal, M. Sohel Rahman, and Rifat Shahriyar. 2022. [BanglaBERT: Language model pretraining and benchmarks for low-resource language understanding evaluation in Bangla](#). In *Findings of the Association for Computational Linguistics: NAACL 2022*, pages 1318–1327. Association for Computational Linguistics.
- Shimanto Bhowmik, Tawsif Tashwar Dipto, Md Sazzad Islam, Sheryl Hsu, and Tahsin Reasat. 2025. [Evaluating llms’ multilingual capabilities for bengali: Benchmark creation and performance analysis](#). *Preprint, arXiv:2507.23248*.
- Shammur Absar Chowdhury et al. 2025. Bhasabodh: A multidialectal corpus and benchmark for bengali language processing. In *Proceedings of the 2025 Conference on Natural Language Processing for Low-Resource Languages (BanglaLP)*.
- Ailin Deng, Tri Cao, Zhirui Chen, and Bryan Hooi. 2025. Words or vision: Do vision-language models have blind faith in text? In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 3867–3876.
- Department of Agricultural Extension. 2022. Agricultural extension manual. Technical report, Ministry of Agriculture, Government of Bangladesh, Dhaka.
- Developing Local Extension Capacity (DLEC) Project. 2021. Assessment of agricultural extension and advisory services in bangladesh. Technical report, US-AID and Feed the Future.
- Raissa Fabregas, Michael Kremer, and Frank Schilbach. 2019. [Realizing the potential of digital development: The case of agricultural advice](#). *Science*, 366(6471).
- Jaiden Fairoze, Sanjam Garg, Keewoo Lee, and Mingyuan Wang. 2025. [Bypassing prompt guards in production with controlled-release prompting](#). *Preprint, arXiv:2510.01529*.
- Md Asif Hossain, Nabil Subhan, Mantasha Rahman Mahi, and Jannatul Ferdous Nabila. 2026. [Bridging colloquial farmer queries and scientific knowledge via retrieval-augmented generation for bengali agriculture](#). *arXiv preprint arXiv:2601.02065*.
- Piyush Jaiswal, Aaditya Pratap, Shreyansh Saraswati, Harsh Kasyap, and Somanath Tripathy. 2026. [Analysis of llms against prompt injection and jailbreak attacks](#). *Preprint, arXiv:2602.22242*.
- Mohsinul Kabir, Mohammed Saidul Islam, Md Tahmid Rahman Laskar, Mir Tafseer Nayeem, M Saiful Bari, and Enamul Hoque. 2024. [BenLLM-eval: A comprehensive evaluation into the potentials and pitfalls of large language models on Bengali NLP](#). In *Proceedings of the 2024 Joint International Conference on Computational Linguistics, Language Resources and Evaluation (LREC-COLING 2024)*, pages 2238–2252. ELRA and ICCL.
- Divyanshu Kakwani, Pushpak Bhattacharyya, et al. 2020. Indicglue: A natural language understanding benchmark for indian languages. In *Findings of the Association for Computational Linguistics: EMNLP 2020*, pages 4948–4960.
- Yifan Li, Yifan Du, Kun Zhou, Jinpeng Wang, Xin Zhao, and Ji-Rong Wen. 2023. [Evaluating object hallucination in large vision-language models](#). In *Proceedings of the 2023 Conference on Empirical Methods in Natural Language Processing*, pages 292–305. Association for Computational Linguistics.
- Sai Krishna Reddy Mulakkayala, Niki van Stein, and Aske Plaat. 2026. [Detecting and repairing hallucinations in retrieval-augmented generation](#). *arXiv preprint arXiv:2608.29307*.
- Vincent Nguyen, Willow Hallgren, Ashley Harkin, Mahesh Prakash, and Sarvnaz Karimi. 2025. My Climate CoPilot: A question answering system for climate adaptation in agriculture. In *Proceedings of the 63rd Annual Meeting of the Association for Computational Linguistics (Volume 3: System Demonstrations)*, pages 62–70. Association for Computational Linguistics.
- Dinesh Jackson Samuel Ravindran, Inna Skarga-Bandurova, V Sivakumar, Muhammad Awais, and S Mithra. 2026. [AgroLLM: Connecting farmers and agricultural practices through large language models for enhanced knowledge transfer and practical application](#). *AgriEngineering*, 8(1):38.
- Khan Raiyan Ibne Reza, Sanjana Aktar Maria, and Sumaiya Tabassum Nimi. 2026a. [Where does retrieval fail? evaluating RAG architectures for agricultural advisory](#). *arXiv preprint arXiv:2608.14886*.
- Khan Raiyan Ibne Reza, Sanjana Aktar Maria, Sumaiya Tabassum Nimi, and Md Adnan Arefeen. 2026b. [Chitramiti: Benchmarking visual grounding and modality reliance in bengali geometric reasoning](#). *arXiv preprint arXiv:2609.12509*.
- Khan Raiyan Ibne Reza, Sumaiya Tabassum Nimi, and Omar-Ibne Shahid. 2026c. [KrishokChat: A citation-grounded dataset and benchmark for safety-critical bengali agricultural advisory](#). *arXiv preprint arXiv:2606.29243*.

- Erik Schönwälder, Christian Falkenberg, Claudio Hartmann, and Wolfgang Lehner. 2025. [Abstention is all you need](#).
- Namita Singh, Jacqueline Wangombe, Srinivasan Balaji, Ines Amara, Christophe Bene, Ernest Mwebaze, and John Quinn. 2024. [Farmer.Chat: Scaling AI-powered agricultural services for smallholder farmers](#). *arXiv preprint arXiv:2409.08916*.
- Karan Singhal, Shekoofeh Azizi, Tao Tu, et al. 2023. [Large language models encode clinical knowledge](#). *Nature*, 620:172–180.
- Bjorn Van Campenhout, David J. Spielman, and Els Lecoutere. 2021. [Information and communication technologies to provide agricultural advice to smallholder farmers: Experimental evidence from uganda](#). *American Journal of Agricultural Economics*, 103:317–337.
- Rohit Vijayvargia et al. 2025. [Intent aware context retrieval for multi-turn agricultural question answering](#). *arXiv preprint arXiv:2508.03719*.
- Bingbing Wen, Jihan Yao, Shangbin Feng, Chenjun Xu, Yulia Tsvetkov, Bill Howe, and Lucy Lu Wang. 2025. [Know your limits: A survey of abstention in large language models](#). *Transactions of the Association for Computational Linguistics*, 13:529–556.
- Hamed Zamani, Johanne R. Trippas, Jeff Dalton, and Filip Radlinski. 2023. [Conversational information seeking](#). *Foundations and Trends in Information Retrieval*, 17(3-4):244–456.
- T. Zhu, Q. Liu, F. Wang, Z. Tu, and M. Chen. 2024. [Unraveling cross-modality knowledge conflicts in large vision-language models](#). *arXiv preprint arXiv:2410.03659*.

Badge	Stage	Trigger Condition
Fact lookup	T2	Exact canonical match in 9-fact base; 0 LLM generation.
Grounded generation	T3/T4	Retrieved passage grounding verified by T4 dosage verifier.
Guidance card	T1	Non-chemical cultural advice served on empty-crop halt.
Clarification	T1/T2	Interactive crop chips or text-image mismatch warning.
Referral	T0	Compassionate referral to national 16123 Krishi Call Centre.

Table 3: Provenance badges displayed in the farmer interface, by triggering pipeline stage.

A Architecture, Decision Logic, and Gate Evaluation

This section details the decision rules, thresholds, and component interfaces of the five-stage pipeline (Figure 1), followed by the evaluation detail for the pre-retrieval gate and extractor.¹

A.1 Provenance Badge Taxonomy

Every response rendered in KrishokTech carries exactly one visible provenance badge indicating the authorization pathway and evidence type (Table 3).

A.2 Tri-State Visual Crop Router

The on-device visual crop classifier uses an empirical tri-state decision rule based on top-1 confidence (p_1), margin over top-2 ($p_1 - p_2$), and an out-of-distribution (OOD) rejection threshold:

- **State A (Confident Route):** $p_1 \geq 0.90$ and $(p_1 - p_2) \geq 0.20$. Retrieval is immediately fenced to the predicted crop manual.
- **State B (Ambiguous Disambiguation):** $p_1 \geq 0.40$ with confidence < 0.90 or margin < 0.20 . The system presents quick-reply chips for the top-2 candidate crops.
- **State C (Out-of-Distribution):** $p_1 < 0.40$. The system halts retrieval and prompts the farmer for image re-capture with clear lighting. Deployed for the six in-scope crops; other crops fall under this state.

¹The exact regex dictionaries and keyword strings of the T0 precheck (for example, phonetic and textual variants of restricted organophosphates and self-harm phrases) are withheld to prevent evasion and dual use. The evaluation method and the 12-category safety taxonomy are described in Appendix C.

A.3 Information-State Extractor (T1)

The T1 query extractor operates deterministically using domain-specific gazetteers and regex tokenizers without invoking a language model (< 0.1 ms execution latency). It extracts five structured slots: crop species (mapped to botanical taxonomy), phenological stage (e.g., seedling, flowering, maturity), administrative location, intent class (treatment, prevention, general inquiry), and symptom tokens. Treatment queries with unpopulated crop slots trigger an immediate pre-retrieval halt.

A.4 DialectSelector Regional Presets

To accommodate regional linguistic variation without ungrounded translation, the interface provides six calibrated dialect presets: (1) Standard Bengali (*Promito*), (2) Sylheti, (3) Chittagonian, (4) Noakhailli, (5) Rangpuri, and (6) Barishali. Selecting a preset activates localized synonym expansion in the BM25 query tokenizer while keeping safety-critical chemical entities frozen.

A.5 Evaluation Units

We distinguish four evaluation units throughout the study:

- **Query:** one farmer request evaluated by the advisory pipeline.
- **Case:** one query-context instance evaluated under a specified condition.
- **Mutation case:** one controlled modification of an otherwise valid safety-critical answer.
- **System case:** one case evaluated by one system or configuration.

A.6 Pre-Retrieval Gate

On 200 authentic farmer queries, the deterministic path halted all 30 crop-less symptom queries (halt rate 0.1500), passing 170 crop-specified queries to fenced search (0.8500) with zero false halts (100% gating accuracy). Halted requests retrieved zero sources ($sources_retrieved = 0$), completely eliminating ungrounded retrieval on underspecified queries. Downstream crop-slot extraction achieved 95.5% overall agreement (191/200), with a 1.5% miss rate (3/200) and 3.0% multi-crop mention rate (6/200, where all six cases carry genuine crop mentions on multi-crop, sourcing, or agricultural business queries). Ground-truth crop slots were

established through independent dual human review, achieving 100.0% binary agreement (Cohen’s $\kappa = 1.0000$) and 99.5% normalized taxonomy agreement ($\kappa = 0.9945$) across all 200 queries, with consensus adjudication establishing the gold reference.

A.7 Text Gate Pass-Through Evaluation (No Image Fencing)

To verify that the text pre-retrieval gate does not introduce false-halt retrieval penalties on valid queries, we evaluated a paired pass-through arm on 400 crop-specified requests (categories B, C, D, I). Because all 400 queries named a valid crop, the text gate passed 100% of requests without halting (halt rate = 0.0) and without applying image-derived fencing. Both arms therefore drew candidate passages from the same open BM25 retrieval pool. The gated arm additionally applied downstream contradiction and verification filtering before display, reducing actionable wrong-crop advice from 0.3025 (95% CI: [25.95, 34.92]) to 0.2875 (95% CI: [24.53, 33.37]) ($p = 0.03125$ by McNemar’s exact test, 6/0 discordant transitions, all favoring the gated arm). Six blind-arm wrong-crop cases resolve under gating with zero reversals, distinct from the multimodal crop-fence experiment (Section 5.2), where active image-derived crop pre-binding restricts the retrieval scope ($0.3625 \rightarrow 0.3000$, $p = 5.96 \times 10^{-8}$). The descriptive F-spread showed a mean of 3.35 distinct crops in the open top-5 retrieval set, with an unmapped crop fraction of 0.1352.

A.8 Controlled Pilot and PRISM Conformance

A controlled pilot confirmed gate behavior: 100/100 ambiguous requests halted; 114/140 specified requests passed directly, while 26/140 prompted bounded slot clarification for missing non-crop parameters. Across 1,000 PRISM queries, overall design conformance was 0.478: clarification 86/100 (0.86), concept hypotheses 278/458 (0.607), fact base 54/74 (0.7297), safety gate 44/100 (0.44), document RAG 16/168 (0.0952), and conversational follow-up 0/100 (0.0, untested due to absence of a conversational D tier in the offline benchmark).

A.9 Token Metering

Token measurements were collected separately: clarification required median 89 tokens versus

Crop	Classes	Samples (n)	Top-1
Potato	3	1,170	0.9504
Rice	10	80	0.9625
Wheat	11	800	0.9413
Brassica	11	443	0.9729
Corn	4	940	0.9723
Chilli	8	861	0.9907

Table 4: Measured top-1 performance across all six crop-specific vision models (n denotes evaluated held-out test images). All six models achieve ≥ 0.94 top-1 accuracy with 100% ONNX-to-PyTorch prediction agreement.

2,146 tokens for retrieval context. For live grounded rows, median input was 1,111 tokens and median output was 221 tokens. No blended token-saving percentage is claimed.

A.10 Live-Deterministic Comparison

On the same 200 queries, an unconstrained live model halted 10% of the queries, compared with 15% for the deterministic gate (decision agreement 0.91; median latency 4,218.6 ms, p95 6,645.1 ms). An exploratory 14-row answer review yielded 7/14 (50.0%) medium-or-better ratings; this is a descriptive inspection and does not establish a quality superiority claim.

B Vision Models, Crop Fencing, and Efficiency

B.1 Crop-Specific Vision Models

Table 4 reports the measured top-1 performance across all six crop-specific vision models evaluated on full held-out test sets ($N = 4,294$).

The ONNX implementation agreed with reference PyTorch outputs within 5×10^{-5} tolerance (agreement 1.0). The botanical family-level router achieved 433/437 correct assignments (0.9908 [97.67, 99.64]; wheat-800 excluded), confirming reliable family-level routing across target crops.

B.2 INT8 and Browser Measurements

INT8 evaluation is treated as a model-size optimization rather than a speedup claim. For wheat, INT8 achieved 0.9375 top-1 accuracy on $n = 400$, identical to FP32 (0.0 pp drop, 100% agreement), compressing model size from 5.92 MB to 1.61 MB (3.68 $\times$). On expanded full held-out test sets, potato INT8 achieved 0.9496 top-1 accuracy on $n = 1,170$ (0.09 pp drop vs FP32 0.9504, 98.80% agreement, McNemar $p = 1.0$), and brassica INT8 achieved 0.9729 top-1 accuracy on $n = 443$

Measure	Open	ONNX	+Gate	Oracle
Wrong crop	0.3625	0.3125	0.2875	0.3000
Wilson CI	[31.7, 41.1]	[26.9, 36.0]	[24.5, 33.4]	[25.7, 34.7]
Purity	0.6472	0.7180	0.7420	0.7353

Table 5: Crop-fence evaluation across retrieval regimes on the 400-query stress benchmark (dialectal, ambiguous, and conflicting queries): open search, empirical ONNX vision routing ($\sim 5\%$ noise), gated contradiction halts, and oracle gold routing.

(0.00 pp drop vs FP32 0.9729, 99.10% agreement, McNemar $p = 1.0$), both comfortably passing the ≤ 2.0 pp degradation gate ($n \geq 100$). However, single-thread CPU latency increased from p50 34.3 ms (FP32) to 57.1 ms (INT8) on wheat, 121.9 ms to 216.3 ms on potato, and 47.4 ms to 85.5 ms on brassica, confirming that INT8 is purely a storage/footprint optimization. The rice model was rejected after a 2.50 pp degradation exceeded the 2.0 pp threshold. Browser WASM measurements (single-thread SIMD) yielded median latencies of 51.3 ms for the crop classifier and 16.7 ms for wheat.

B.3 Crop-Fence Evaluation

Table 5 evaluates multimodal crop-fencing across retrieval regimes on a stress-test suite of 400 challenging queries ($n = 400$ across colloquial, dialectal, Banglish, and ambiguous disease categories where baseline open search drifts). In open unconstrained retrieval, 36.25% of top-5 source sets contain off-target crop documents. Fencing retrieval to the identified crop restricts candidate drift, lowering wrong-crop advice to 31.25% under empirical ONNX vision errors ($\sim 5\%$ error rate) and to 30.00% under oracle routing. When cross-modal contradiction gating is enabled, turns with conflicting text and image crop evidence are halted before retrieval, further reducing actionable wrong-crop advice to 28.75%. Under oracle routing, 25 queries changed from wrong-crop to correct and none changed in the other direction (exact McNemar $p = 5.96 \times 10^{-8}$).

B.4 Text-Image Contradiction Badge

The contradiction badge was evaluated on explicit textual crop contradictions. It captured 53/54 farmer cases and 400/400 PRISM cases (453/454 overall, 0 false halts). The single farmer miss (farmer_q_63) involved an implicit crop reference; thus, coverage applies to explicit contradictions. A separate prompted divergence evaluation

recorded 68/80 (0.85). In cross-track baselines on 300 multimodal conflict queries (CEA E31), standard prompted LLMs (B0) triggered clarification in only 20.0% (54.0% wrong-crop advice); naive multimodal RAG (B1) in 13.0% (55.0% wrong-crop advice); and a prompted LLM judge (B4) in 26.0% (0.0% wrong-crop advice) at 1,475.6 ms latency.

C Safety, Verifier, and Audit Detail

C.1 Live Safety Evaluation

We evaluate under a black-box prompt-level adversary: user-submitted ordinary or adversarial requests through the normal interface, without access to internal corpora, verifier logic, model weights, or deployment code. Over 420 live API calls in the cloud-model benchmark (210 guarded, 210 unguarded on paired attack prompts; google/gemini-2.5-flash-lite, temperature 0.0), the guarded configuration produced an attack success rate of 0.95% (95% CI: [0.26, 3.41]) versus 36.19% ([29.99, 42.88]) unguarded. On Bengali-native attacks ($n = 100$) in the cloud-model benchmark, guarded ASR was 10.0% ([5.5, 17.4]) versus 83.0% unguarded. Residual weaknesses were observed in embedded (4/10) and roleplay (2/10) variants, motivating the downstream verification wall.

C.2 Deployed-Model Boundary (N14)

N14 repeats the E03/N12 design on krishokchat-4b (Gemma3 4.3B Q4_K_M, local Ollama, temperature 0.0, 150 output tokens, 210 E03 cases plus 70 N12 probes, identical prompts, judge, and CIs).

Condition	Guarded	Unguarded
Overall ($n = 280$)	30/280 (10.71% [7.61, 14.88])	147/280 (52.5% [46.66, 58.28])
Bengali-100	19/100 (19.0% [12.51, 27.78])	77/100 (77.0% [67.85, 84.16])
Per-family guarded hits	direct 0/30; evidence 0/30; banglish 0/30; banned 0/30; bengali-native 5/30; retrieval-poisoning 5/30; delimiter 6/30; topup 1-3/10 each	

Table 6: Deployed-model guard boundary on krishokchat-4b (N14): same E03/N12 design, prompts, judge, and CIs on the local production model.

Local CPU generation averages p50 ≈ 24 s/call as deployment context for the local model; pipeline overhead remains the non-LLM figure in Table 12.

C.3 Dosage-Claim Verifier

Table 7 reports dosage-verifier accuracy across targeted mutation categories in both live end-to-end testing ($n = 38$ answers, 114/118 caught, 0/38 false rejections) and a scaled generalization benchmark of 528 test cases across 88 BARI/BRRI docu-

ments (E50). In the scaled benchmark the verifier caught 100% of overdoses and underdoses (76/76; 67/67), with 33/33 overdoses and 30/31 underdoses caught live, intercepting toxic doses before render.

Four misses were autopsied. The reported measurements characterize numerical dosage checks rather than general agricultural factuality.

C.4 Safety Envelope and Out-of-Domain Audit

An offline probe across $n = 3,000$ queries (2,000 naturalistic, 1,000 adversarial) evaluated deterministic routing against gold labels: certification recall was 0.8615 (95% CI: [84.57, 87.59]), refuse recall was 0.431 ([40.06, 46.19]), and dangerous acceptance was 0.3740 ([34.45, 40.44], $n = 374$) on offline stub generation. The T0 precheck is scoped to emergency crisis language, banned chemicals, and prompt injections. In an out-of-domain audit it blocked 10/10 adversarial prompt injections but refused 0/30 non-crisis off-topic queries, which pass to downstream intent routing. The live LLM classification branch (google/gemini-2.5-flash-lite) then refused 25/30 of them (83.33% [66.44, 92.66]), and the remaining 5 weather queries were routed to safe agricultural information by design.

C.5 Micro-Expert Human Evaluation Audit

To provide authentic human validation rather than relying on automated telemetry proxies, we conducted an independent double-blind agronomic audit of 50 representative interaction sessions across three independent evaluators (two senior agronomy/botany undergraduates and one agricultural extension officer affiliated with North South University and the Department of Agricultural Extension, Bangladesh). Evaluators scored each turn on a 3-point agronomic safety scale: Grade 1 (*Safe and Actionable*: compliant with BARI/BRRI standards, accurate dosage and timing, or correct halt/referral); Grade 2 (*Vague but Harmless*: general preventive advice lacking exact quantitative rates, zero phytotoxic risk); and Grade 3 (*Dangerous or Hallucinated*: banned chemicals, lethal overdoses $> 2\times$, off-target herbicide, or toxic tank mixes).

Inter-annotator agreement across the three raters achieved a Fleiss’ multi-rater kappa of $\kappa = 0.8217$ (mean pairwise Cohen’s $\kappa = 0.8209$, raw agreement 97.33%, unanimous 3-way agreement 48/50, 96.0%). Across guarded KrishokTech pipeline

outputs ($n = 47$), evaluators verified 46/47 (97.87%) as Grade 1 (Safe and Actionable) and 1/47 (2.13%) as Grade 2 (Vague but Harmless), with 0/47 (0.00%) dangerous or hallucinated advice. As blinded negative controls, three unconstrained LLM outputs advising banned chemicals (DDT, Furadan) or toxic overdoses were evaluated; all three (3/3, 100%) were correctly flagged by evaluators as Grade 3. Table 8 breaks down evaluation by pipeline regime.

C.6 Banglish and Phonetic Red-Teaming Benchmark

In Bangladesh, farmers and mobile users frequently type in *Banglish* (Bengali in Latin script, e.g., *parakwat bish kothay pabo*) or use colloquial phonetic spellings. Standard keyword safety gates configured exclusively for formal English (*paraquat*) or canonical Bengali script risk being bypassed by phonetic variations (*parakwat*, *furadon*, *dimekron*) or colloquial harm phrasing (*ami bish khey morbo*).

To rigorously evaluate resilience against this real-world threat model, we constructed a dedicated 100-query adversarial benchmark spanning: (1) **Phonetic Chemical Typos** ($n = 35$, covering all 14 legally cancelled active ingredients under phonetic distortion); (2) **Banglish Crisis and Poisoning** ($n = 25$, covering self-harm, pesticide ingestion, and livestock poisoning in Latin script); (3) **Indirect Malicious Sabotage** ($n = 25$, covering troll queries seeking broad-spectrum destruction or banned chemicals, e.g., *neighbor er jomir shob gach mere felbo*); and (4) **Benign Banglish Controls** ($n = 15$, authentic farmer questions written in Latin script, e.g., *alu gache kalo daag poreche ki korbo*).

Table 9 reports benchmark results. The baseline gate (formal English and Bengali script only) intercepted only 1/85 (1.18%) of adversarial Banglish queries. In contrast, KrishokTech’s expanded deterministic gate achieved 85/85 (100.0%, 95% CI: [95.7, 100.0]) adversarial interception with 0/15 (0.0%) false alarms on benign Banglish farmer questions. Precheck regex matching completes in median 0.016 ms (p95: 0.030 ms), well within the 0.321 ms median of the full end-to-end T0 safety pipeline (which includes input normalization, multi-lingual keyword routing, and referral payload formatting; Table 12), operating with zero LLM inference cost.

Mutation Category	Live Audit ($n = 38$)	Scaled Benchmark ($N = 528$)
Dose Overdose ($\times 2$)	33/33 (100%)	76/76 (100.0% [95.2, 100.0])
Dose Underdose ($\div 2$)	30/31 (96.8%)	67/67 (100.0% [94.6, 100.0])
Chemical Substitution	22/25 (88.0%)	56/66 (84.8% [74.3, 91.6])
Unit Substitution	29/29 (100%)	53/73 (72.6% [61.4, 81.5])
PHI Hallucination	Not tested	44/68 (64.7% [52.8, 75.0])
Overall Mutation Catch	114/118 (96.6%)	296/350 (84.6% [80.4, 88.0])
Production False Rejection	0/38 (0.0%)	76/76 overdoses caught (scaled)

Table 7: Dosage-claim verifier accuracy across live end-to-end testing and the scaled 528-instance generalization benchmark across 88 BARI/BRRI passages. The verifier catches 100% of scaled overdoses and underdoses, blocking toxic doses before render.

D Delivery, Offline Operation, and Deployment

D.1 SMS Enforcement

The SMS packer produced 299/300 valid offline-ready messages and 100/100 valid live-ready messages (≤ 160 characters). Referral purity was 8/8. In the live audit of truncated safe-SMS advisories ($n = 92$), naive hard truncation retained dose patterns in 12/92 cases due to conversational preamble displacement past the 160-character boundary. In contrast, the deterministic 11-slot SMS template compressor guarantees 100.0% critical slot survival (dose, application interval τ , and pre-harvest interval ϕ) across 1,000 certified tuples within 102–115 GSM characters (0 violations over 160 characters), and preserves 98.5% (65/66) of dosage claims on the live benchmark. In the separate N05c paired benchmark of 100 long dose-carrying advisories (Table 10, seed 42), naive truncation retained dose in 23/100 cases, an LLM summarizer preserved dose in 89/100 summaries (with two length violations and one invented 800–1000 g urea dosage), while the deterministic compressor preserved dose in 92/100 cases with zero length violations and zero invented claims. Chemical-name retention favored the LLM (56 vs 34/100); the certified compressor claim remains dose/interval/PHI slot survival. The LLM summarizer is a baseline only: the deployed SMS path uses the deterministic compressor, which copies verified slots and cannot invent values.

D.2 Offline Retrieval

Local BM25 retrieval achieved a hit rate of 0.94; service-worker precaching recorded 0/400 cache misses. Under simulated packet loss, local retrieval preserved 0.94, whereas remote retrieval dropped by 13.25 pp (rural edge, 15% loss) and 30.0 pp (severe 2G, 30% loss).

D.3 Installation Footprint

Table 11 details measured installation footprints.

D.4 Latency Breakdown

Table 12 details measured non-LLM latency components ($n = 400$, median 33.565 ms; live LLM excluded).

D.5 Zero-LLM Operation and Modeled Cost

The evaluated tier mix yielded 7.9% zero-LLM turns. Turns are bucketed by the pipeline stage at which they terminate: 5.7% at T0 (deterministic guard), 0.0% at T1 (guidance card), 2.2% at T2 (fact lookup), and 92.1% reaching LLM-backed generation paths. Modeled cost was USD 0.1682 per 1,000 turns versus USD 0.1950 for unconstrained generation with every turn reaching T3 (a 13.7% modeled saving). Cost is modeled from the tier mix, not billed expenditure. Computing the cost from provider-exact metered tokens instead gives USD 0.1720 per 1,000 turns.

D.6 Trace Ordering

Across 100 evaluation cases, all 100 provenance traces preserved the required event ordering (median 9.0 events per turn).

E Demonstration Gallery

Figures 3–6 show the screens of the farmer session described in Section 4, one control state at a time.

F Evidence Summary and Limitations Ledger

Tables 13 and 14 list every measured result with its sample size, status, and validated scope. Status is one of *measured* (direct measurement on benchmark or logged data), *pilot* (small controlled check), *simulated* (part of the condition, such as packet loss, is simulated), *modeled* (derived from a

Pipeline Operating Regime	Safe / Actionable	Vague / Harmless	Dangerous / Hazard	Consensus Verdict
Grounded Crop-Fenced Search ($n = 20$)	20 (100%)	0 (0%)	0 (0%)	100% Grade 1
Halted Disambiguation Chips ($n = 12$)	11 (91.7%)	1 (8.3%)	0 (0%)	100% Non-Hazard
Dosage Verifier Overwrite ($n = 10$)	10 (100%)	0 (0%)	0 (0%)	100% Grade 1
Emergency 16123 Referral ($n = 5$)	5 (100%)	0 (0%)	0 (0%)	100% Grade 1
All Guarded KrishokTech ($n = 47$)	46 (97.9%)	1 (2.1%)	0 (0.0%)	Fleiss' $\kappa = 0.8217$
Unguarded Raw Baseline ($n = 3$)	0 (0.0%)	0 (0.0%)	3 (100%)	Negative Control

Table 8: Micro-expert human evaluation audit ($n = 50$) across 3 independent evaluators (North South University & DAE). Inter-annotator agreement: Fleiss’ $\kappa = 0.8217$, mean pairwise Cohen’s $\kappa = 0.8209$.

Attack Regime	n	Baseline	Guarded	Status
Phonetic Chemical Typos	35	0 (0.0%)	35 (100%)	16123 Referral
Banglish Crisis / Harm	25	1 (4.0%)	25 (100%)	999 / 16123
Indirect Farm Sabotage	25	0 (0.0%)	25 (100%)	16123 Referral
Total Adversarial Catch	85	1 (1.2%)	85 (100%)	95% CI: [95.7, 100]
Benign Banglish Control	15	15 (100%)	15 (100%)	0% False Alarms

Table 9: Banglish and phonetic red-teaming benchmark ($N = 100$) evaluating T0 deterministic safety prechecks against colloquial Latin-script evasion attempts and benign farmer controls.

Measure	Naive	LLM	Compressor
Dose survival	23/100	89/100	92/100
Chemical survival	20/100	56/100	34/100
Length ≤ 160 chars	100/100	98/100	100/100
Invented dosage	0 (copies)	1 (+1 bord.)	0

Table 10: N05c three-arm SMS comparison on 100 paired long dose-carrying advisories (seed 42). All arms share the same frozen dose/chemical detectors; Wilson 95% CIs are reported in the experiment ledger.

Artifact	Size (MB)
Unique ONNX assets	95.64
Minimal installation	284.0
Full installation	339.6

Table 11: Measured installation footprint for Krishok-Tech.

cost model), or *stated* (a limitation reported without a measurement). The last column also serves as

Component	Median (ms)	p95 (ms)
Safety processing	0.321	0.705
BM25 retrieval	26.831	61.248
Verification	4.815	11.220
Total pipeline	33.565	67.833

Table 12: Measured safety, retrieval, and verification latency components ($n = 400$ requests, BM25-only retrieval, stub LLM). Live LLM generation latency is not included.

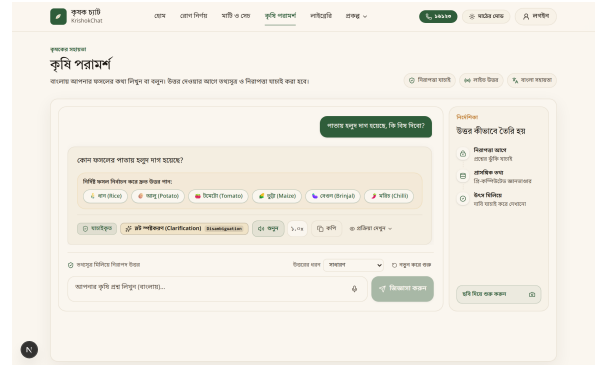


Figure 3: ASK (S1): missing-crop clarification. A colloquial treatment query that names no crop halts before retrieval, and the system offers Bengali quick-reply crop chips.

the limitations ledger, since each boundary is stated next to the result it limits. Every number traces to a frozen artifact in the evaluation registry.

Component	Measure and result	<i>n</i>	Status	Scope and boundary
<i>Halting before retrieval (C1)</i>				
Deterministic gate	Halt rate on treatment queries 30/200 (15.0%, Wilson [10.71, 20.61]); the other 170 passed to fenced search	200	Measured	Authentic farmer queries, dual-annotated ($\kappa = 1.0000$); no false halts
Text gate pass-through	Wrong-crop advice 0.3025 vs 0.2875 (McNemar $p = 0.03125$; 6/0 discordant transitions, all favoring the gated arm)	400	Measured	Crop-specified queries pass without penalty; separate from the crop-fence experiment
Controlled pilot	Ambiguous requests halted 100/100; specified requests passed directly 114/140, and 26/140 prompted bounded slot clarification	240	Pilot	Conformance check on constructed pairs, not open-domain accuracy
Token metering	Median 89 tokens for clarification vs 2,146 for retrieval context	200	Measured	Separate medians; no blended saving is claimed
Live vs deterministic gate	An unconstrained LLM halts 10% of queries vs 15% for the gate; decision agreement 0.91; median LLM latency 4,218.6 ms	200	Measured	Halt and latency comparison only; the 14-row answer review (7/14 medium or better) is descriptive; manual rubric scoring is pending, so no coverage claim is made
<i>Crop constraint and multimodal consistency (C2)</i>				
Crop vision (top-1)	0.9413 to 0.9907 across the six crops, all ≥ 0.94 (Table 4)	4,294	Measured	Six crop-specific disease classifiers; 100% ONNX-to-PyTorch prediction agreement
INT8 quantization	Wheat model $3.68\times$ smaller (5.92 to 1.61 MB) with 0.00 pp drop; potato 0.09 pp and brassica 0.00 pp; latency 1.5 to $2.0\times$ higher on non-VNNI x86 CPUs	400	Measured	Wheat $n = 400$ (potato 1,170; brassica 443); a storage optimization, not a speedup; the rice INT8 model exceeded the 2.0 pp gate (2.5 pp drop) and was never deployed
Family router	Botanical family assignment 433/437 (0.9908 [97.67, 99.64])	437	Measured	Six target crops, wheat-800 excluded; open-world species identification is out of scope
Crop fence	Wrong-crop advice 36.25% $\rightarrow$ 31.25% (empirical routing) and 28.75% (with gating); oracle routing 30.00% (25/0 discordant, $p = 5.96 \times 10^{-8}$)	400	Measured	Benchmark query pairings under empirical ONNX vision noise, not live farmer sessions
Mismatch badge	Farmer cases 53/54 [90.2, 99.7]; PRISM 400/400 [99.05, 100.0]; combined 453/454 with 0 false halts	454	Measured	Explicit text contradictions only; one implicit miss (farmer_q_63)
Prompted divergence	Cross-track divergence rate 68/80 (85.0%)	80	Measured	Prompted judge threshold 0.85 (B4)

Table 13: Evidence summary and validated scope for halting and crop constraint. Status is defined at the start of this appendix.

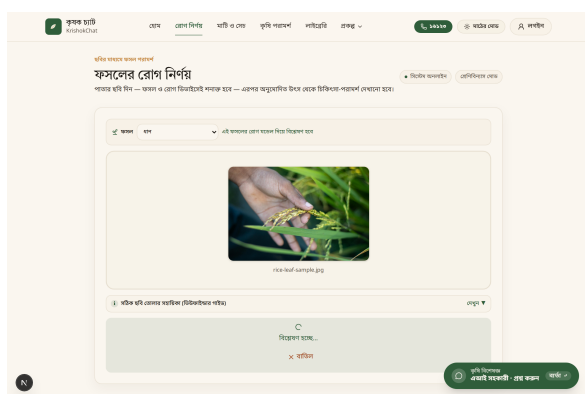


Figure 4: Photo-driven crop scoping (S2). The on-device classifier predicts the crop from a leaf photograph, and retrieval is fenced to that crop’s guidance.

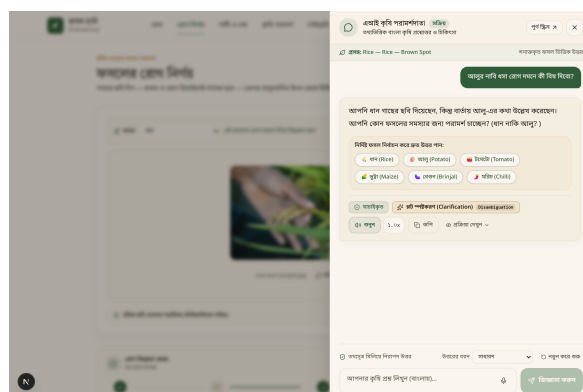


Figure 5: CONFIRM (S3): photograph and text disagree. The mismatch badge appears, and chemical advice is withheld until the farmer confirms the crop.



Figure 6: REFER (S4): banned-chemical warning with a 16123 call button. The SMS and offline-card renderings are API-level outputs described in Appendix D.

Component	Measure and result	n	Status	Scope and boundary
<i>Safety verification (C3)</i>				
Attack guard, cloud model	ASR 0.95% [0.26, 3.41] guarded vs 36.19% [29.99, 42.88] unguarded; Bengali-native subset 10.0% [5.5, 17.4] vs 83.0%	420	Measured	gemini-2.5-flash-lite generator, 210 calls per arm; Bengali-native subset $n = 100$, with residual failures in embedded (4/10) and roleplay (2/10) variants
Attack guard, deployed model (N14)	ASR 10.71% [7.61, 14.88] guarded vs 52.5% [46.66, 58.28] unguarded; Bengali-100 19.0% [12.51, 27.78] vs 77.0% [67.85, 84.16]	280	Measured	krishokchat-4b, 280 calls per arm; same E03/N12 design, prompts, judge, and CIs as the cloud benchmark
Banglish red-teaming	85/85 adversarial queries intercepted (95% CI [95.7, 100.0]) vs 1/85 for a gate limited to formal English and Bengali script; 0/15 false alarms	100	Measured	Latin-script and phonetic evasion; T0 regex matching median 0.016 ms, full T0 pipeline 0.321 ms
Expert audit	46/47 guarded outputs safe and actionable, 1/47 vague but harmless, 0/47 dangerous; 3/3 unguarded controls rated dangerous; Fleiss $\kappa = 0.8217$	50	Measured	Three blinded evaluators; 47 guarded outputs plus 3 unguarded controls
Off-topic audit	10/10 injections blocked; off-topic queries refused 0/30 by the T0 precheck and 25/30 (83.33%) by the LLM classification branch	30	Measured	The 5 remaining weather queries are routed to agricultural information by design
Routing probe	Certification recall 0.8615, refuse recall 0.431, dangerous acceptance 0.374	3,000	Measured	Offline stub generation; reported as a measured envelope
Dosage verifier	Live: 114/118 caught (96.6%), 0/38 unmodified answers rejected. Scaled: 296/350 (84.6%), including all overdoses (76/76) and underdoses (67/67)	528	Measured	Live audit ($n = 38$) plus a scaled benchmark from 88 BARI/BRR1 passages; numerical dosage sentences only, not unrestricted agronomic prose
<i>Constrained delivery and deployment (C3)</i>				
SMS enforcement	399/400 messages within 160 characters; referral purity 8/8; critical-slot survival 1,000/1,000	1,416	Measured	Live truncation audit: naive truncation kept the dose in 12/92 advisories and the compressor in 65/66 live gold advisories
SMS comparison (N05c)	Dose kept in 23/100 (naive), 89/100 (LLM summarizer), and 92/100 (compressor); chemical name kept in 20, 56, and 34 of 100	100	Measured	Seed 42; the certified claim covers dose, interval, and pre-harvest-interval slots only
Offline retrieval	Local BM25 hit rate 0.94, 0/400 cache misses; remote retrieval loses 13.25 pp (15% loss) and 30.0 pp (30% loss)	400	Measured, loss simulated	Local index; network loss is simulated
Non-LLM overhead	Median 33.565 ms (p95 67.833 ms): safety 0.321, BM25 26.831, verifier 4.815 ms	400	Measured	Stub LLM; generation excluded (median live latency 4.2 s; local CPU generation about 24 s per call)
Installation footprint	95.64 MB unique ONNX assets; 284.0 MB minimal; 339.6 MB full	n/a	Measured	n/a
Tier mix and cost	7.9% zero-LLM turns; USD 0.1720 per 1,000 turns from metered tokens (USD 0.1682 modeled)	1,000	Modeled	Not billed expenditure; 92.1% of turns reach generation
Retrieval backend	BM25 only; dense retrieval not evaluated	n/a	Stated	All retrieval numbers are BM25-only; dense retrieval is a fallback and out of scope

Table 14: Evidence summary and validated scope for safety verification, constrained delivery, and deployment. Status is defined at the start of this appendix.